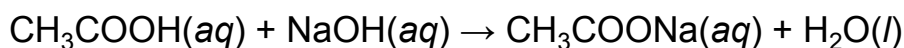


The reaction of CH_3COOH (a weak acid) with NaOH (a strong base) forms CH_3COONa (a salt) and water, as shown below.



If a sample of $\text{CH}_3\text{COONa}(s)$ salt is isolated and dissolved in pure water, the estimated pH of the resulting salt solution is:

- ☐ A. < 4 . [2%]
☐ B. > 4 and < 7 . [20%]
☐ C. $= 7$. [13%]
✓ ☒ D. > 7 . [64%]

Correct

64% answered correctly

Explanation:

Salt formation table

$\text{CH}_3\text{COOH} + \text{NaOH} \rightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O}$

		Ion from:	
		Strong acid	Weak acid
Ion from:	Strong base	Neutral salt ($\text{pH} = 7$)	Basic salt ($\text{pH} > 7$)
	Weak base	Acidic salt ($\text{pH} < 7$)	Conditional ($\text{pH} = ?$)*

*The pH produced by these salts depends upon the relative strength of the weak acid & weak base.

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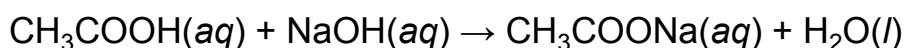
Salts are formed by a **neutralization** reaction between an **acid and a base**. Because the resulting salt is made from a **conjugate acid or base** formed during the neutralization, the salt may react with water to produce OH^- or H_3O^+ (ie, a **hydrolysis** reaction) and affect the pH of the solution.

The tendency of the salt ions to act as an acid or a base depends on the strengths of the acid and the base involved in forming the salt. As such, examining the origins of the ionic species in a salt permits a qualitative assessment of the pH of a salt solution:

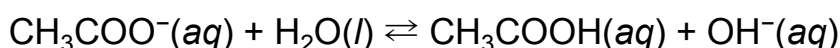
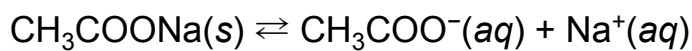
- **Neutral salts** formed from a **strong acid** reacting with a **strong base** give neutral solutions ($\text{pH} = 7$).
- **Acidic salts** formed from a strong acid reacting with a **weak base** give acidic solutions ($\text{pH} < 7$).

- **Basic salts** formed from a **weak acid** reacting with a strong base give basic solutions ($\text{pH} > 7$).
- Salts formed from a weak acid reacting with a weak base are more difficult to broadly classify, and the solution pH of these salts often must be determined experimentally or by calculation.

Although the pH of a 0.001 M CH_3COONa salt solution can be **calculated**, an exact solution pH is not required by the question, and the value can be assessed qualitatively. Accordingly, the component **ions of CH_3COONa** indicate that CH_3COONa is a basic salt produced by the reaction of CH_3COOH (a weak acid) with NaOH (a strong base).



Therefore, a sample of CH_3COONa salt dissolved in water will dissociate into its component ions, and the CH_3COO^- ion will act as a weak base and undergo hydrolysis to form OH^- ions, resulting in a slightly basic solution ($\text{pH} > 7$).



(Choices A, B, and C) CH_3COONa is a basic salt and will not produce an acidic or neutral solution ($\text{pH} \leq 7$) from hydrolysis.

Educational objective:

The pH of a salt solution is determined by the extent of hydrolysis caused by the nature of the salt. Neutral salts yield solutions with $\text{pH} = 7$, acidic salts give solutions with $\text{pH} < 7$, and basic salts produce solutions with $\text{pH} > 7$.

Time spent: 0 sec

Subject:

General Chemistry

QID: 402105

System:

Solutions and Electrochemistry